

2026

Data Mining

Assignment :- 2

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DM-Assignment2.py > ...

```
1  ∨ students = [  
2      {"id": 1, "name": "Shyam", "marks": 85, "attendance": 92},  
3      {"id": 2, "name": "Priya", "marks": 72, "attendance": 85},  
4      {"id": 3, "name": "mukul", "marks": 55, "attendance": 68},  
5      {"id": 4, "name": "Nehal", "marks": 91, "attendance": 95},  
6      {"id": 5, "name": "Ravin", "marks": 45, "attendance": 60},  
7      {"id": 6, "name": "Sneha", "marks": 78, "attendance": 88},  
8      {"id": 7, "name": "Karam", "marks": 62, "attendance": 75},  
9      {"id": 8, "name": "Akash", "marks": 39, "attendance": 55}  
10 ]  
11  
12  
13 # Original Data  
14 print("----- STUDENT DATA -----")  
15 print("ID", "Name", "Marks", "Attendance")  
16 ∨ for s in students:  
17 ∨     print(  
18         s["id"],  
19         s["name"], " ",  
20         s["marks"], " ",  
21         s["attendance"]  
22     )  
23  
24  
25 # 1. TRANSFORMATION - Min-Max Normalization  
26  
27 all_marks = [s["marks"] for s in students]  
28  
29 low = min(all_marks)  
30 high = max(all_marks)  
31  
32 print("\n----- TRANSFORMATION -----")  
33 print("Normalized Marks:")  
34
```



```
35 for s in students:
36     value = (s["marks"] - low) / (high - low)
37     print(s["name"], "=>", round(value, 2))
38
39
40 # 2. AGGREGATION - Total and Average
41
42 sum_marks = 0
43
44 for s in students:
45     sum_marks += s["marks"]
46
47 avg_marks = sum_marks / len(students)
48
49 print("\n----- AGGREGATION -----")
50 print("Total:", sum_marks)
51 print("Average:", round(avg_marks, 2))
52
53
54 # 3. DISCRETIZATION
55
56 print("\n----- DISCRETIZATION -----")
57
58 for s in students:
59
60     score = s["marks"]
61
62     if score < 50:
63         level = "Poor"
64     elif score < 75:
65         level = "Average"
66     else:
67         level = "Good"
68
69     print(s["name"], "=>", level)
```

```
71
72 # 4. BINARIZATION
73
74 print("\n----- BINARIZATION -----")
75
76 for s in students:
77     score = s["marks"]
78
79     if score >= 50:
80         status = 1
81     else:
82         status = 0
83
84     print(s["name"], "=>", status)
85
86
87
88 # 5. SAMPLING
89
90 print("\n----- SAMPLING -----")
91 print("Taking alternate records:")
92
93 for i in range(0, len(students), 2):
94     print(students[i])
```



• (.venv) PS D:\DM assignment 1> python DM-Assignment2.py

----- STUDENT DATA -----

ID	Name	Marks	Attendance
----	------	-------	------------

1	Shyam	85	92
---	-------	----	----

2	Priya	72	85
---	-------	----	----

3	mukul	55	68
---	-------	----	----

4	Nehal	91	95
---	-------	----	----

5	Ravin	45	60
---	-------	----	----

6	Sneha	78	88
---	-------	----	----

7	Karam	62	75
---	-------	----	----

8	Akash	39	55
---	-------	----	----

----- TRANSFORMATION -----

Normalized Marks:

Shyam => 0.88

Priya => 0.63

mukul => 0.31

Nehal => 1.0

Ravin => 0.12

Sneha => 0.75

Karam => 0.44

Akash => 0.0

----- AGGREGATION -----

Total: 527

Average: 65.88

----- DISCRETIZATION -----

Shyam => Good

Priya => Average

mukul => Average

Nehal => Good

Ravin => Poor

Sneha => Good

Karam => Average

Akash => Poor

----- BINARIZATION -----

Shyam => 1

Priya => 1

mukul => 1

Nehal => 1

Ravin => 0

Sneha => 1

Karam => 1

Akash => 0

----- SAMPLING -----

Taking alternate records:

{'id': 1, 'name': 'Shyam', 'marks': 85, 'attendance': 92}

{'id': 3, 'name': 'mukul', 'marks': 55, 'attendance': 68}

{'id': 5, 'name': 'Ravin', 'marks': 45, 'attendance': 60}

{'id': 7, 'name': 'Karam', 'marks': 62, 'attendance': 75}

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THANK
For your attention
YOU

